WHITE PAPER · YCM TV-188 B × AEROSPACE

YCM TV-188 B for Aerospace

Large-envelope CNC vertical mill for oversized workpieces, complex pocketing, and multi-face machining. This paper covers what the machine does, the aerospace parts we produce with it, the alloys we run, the tolerances and finishes aerospace buyers expect, and the documentation package.

Loco Motive · 32" W × 70" L × 32" H workpiece envelope · large-envelope CNC vertical machining center

Executive summary

Machine: YCM TV-188 B (shop nickname: *Loco Motive*). large-envelope CNC vertical machining center at B&R Productions, running from our New Waverly, Texas shop under an ISO 9001:2015 quality system.

Application: Large-envelope CNC vertical mill for oversized workpieces, complex pocketing, and multi-face machining for Aerospace customers — Aerospace primes, tier-1 and tier-2 suppliers, MRO shops, propulsion component manufacturers, structural assembly integrators, actuator OEMs.

Envelope: 32" W × 70" L × 32" H workpiece envelope. **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

The machine, in context

Loco Motive is where the oversized work lives — valve bodies, wellhead spool adapters, large manifold blocks, and any part that busts the 20-inch class of mills. YCM's TV-188 platform is a heavier, more rigid class of vertical machining center; the extra column and table mass matters when you're taking heavy roughing cuts in Inconel or hogging out a wellhead block.

Full spec table

Model	YCM TV-188 B ("Loco Motive")
Type	3-axis CNC vertical machining center — heavy-duty class
Workpiece envelope	32" W × 70" L × 32" H (B&R working spec)
Axis travels	70.9" X × 33.9" Y × 29.5" Z
Table size	78.7" × 33.9" (2,000 × 860 mm)
Max workpiece weight	4,409 lb (2,000 kg)
Spindle taper	BT50
Spindle speed	6,000 RPM standard (10,000 RPM optional)
Spindle power	25 HP standard (30 HP optional), 30-min rating
Automatic tool changer	24 tools standard (40 optional)
Rapid traverse	591 IPM; cutting feed 197 IPM
Control	Fanuc MXP-200i
Structure	Patented T-base for rigidity + chip disposal
Tolerance capability	±0.0005" routine on critical features
Materials	Stainless, carbon steel, tool steel, Inconel, Hastelloy, aluminum, titanium

Who this serves in Aerospace

Typical buyer: Aerospace primes, tier-1 and tier-2 suppliers, MRO shops, propulsion component manufacturers, structural assembly integrators, actuator OEMs.

What's hard about aerospace work: Certification-driven documentation, exotic superalloys, tight tolerances on structural and engine components, ITAR handling for controlled prints, and customer-specific QMS integration.

Typical Aerospace parts we produce on Loco Motive

- Large structural brackets and airframe fittings
- Manifold blocks and valve housings for hydraulic systems
- Actuator housings and pylon components
- Engine mount hardware requiring extended envelope
- Large tooling and fixtures for aerospace assembly

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Aerospace

Standard range: Titanium Grade 5 (Ti-6Al-4V) and Grade 23 (Ti-6Al-4V ELI) · Inconel 718 · Inconel 625 · Waspaloy · A286 · 17-4 PH · 15-5 PH · 13-8 Mo PH · Aluminum 7075/6061 · Custom aerospace alloys on request.

Titanium Grade 5 is the workhorse and it's temperature-sensitive: aggressive chip load with flood coolant keeps the heat in the chip, not the workpiece. Light cuts overheat titanium and cook tools; that's why titanium demands a different intuition than steel. Inconel 718 in aerospace-aged condition is harder and less forgiving than the annealed billet stock — most aerospace 718 work assumes aged. 17-4 PH and 15-5 PH are common for structural and actuator components; both are precipitation-hardening stainless with condition-specific machining plans.

> **What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific

> to each alloy family. Your part is not the one we're experimenting on.

Tolerance and finish expectations in Aerospace

±0.0005" routine on critical features. Tighter with proper fixture and setup — some aerospace features run ±0.0002" or GD&T-driven cylindricity and position tolerances. CMM verification standard on first articles; process-driven measurement on runs. Surface finish measured, not estimated.

Documentation and quality workflow

Customer-specific quality plans integrated on demand. First-article layout and CMM verification standard. Material traceability by heat and lot. ITAR-controlled prints handled under NDA with customer-specific document control — prints stay in-shop. Certificates of conformance issued with delivery.

Response time and emergency work

AOG (aircraft-on-ground) work prioritized when material is available. Emergency response supported for existing aerospace customers with blanket arrangements or established relationships. Call direct at (936) 291-7827.

How we compare

Compared to a captive aerospace shop: we're faster for one-offs and small runs, more flexible on setup, and we don't require a long approval process. Compared to a lowest-bidder commodity shop: we understand the alloys, we've handled ITAR before, and we treat first-article inspection as a discipline rather than a formality.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request. ITAR-controlled prints handled under document control with prints staying in-shop.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Ready to talk about your Aerospace work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

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